Carbon dioxide is prepared in laboratory by the action of dilute hydrochloric acid on metal carbonates.

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

Properties

It is a colourless, nonflammable gas and is heavier than air. Its critical temperature is 31^0 C and can be readily liquefied.

Carbon dioxide is a very stable compound. Even at 3100 K only 76 % decomposes to form carbon monoxide and oxygen. At still higher temperature it decomposes into carbon and oxygen.

$$CO_2 \underset{}{\overset{3100\ K}{\rightleftharpoons}} CO + \tfrac{1}{2}O_2$$

$$CO_2 \overset{\text{high temperature}}{\rightleftharpoons} C + O_2$$

Reducing behaviour:

At elevated temperatures, it acts as a strong reducing agent. For example,

$$CO_2 + Mg \longrightarrow 2MgO + C$$

Water gas equilibrium:

The equilibrium involved in the reaction between carbon dioxide and hydrogen, has many industrial applications and is called water gas equilibrium.

$$CO_2 + H_2 \rightleftharpoons \underset{\text{Water gas}}{CO + H_2O}$$

Acidic behaviour:

The aqueous solution of carbon dioxide is slightly acidic as it forms carbonic acid.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Structure of carbon dioxide

Carbon dioxide has a liner structure with equal bond distance for the both C-O bonds. In this molecule there is one C-O sigma bond. In addition there is 3c-4e bond covering all the three atoms.

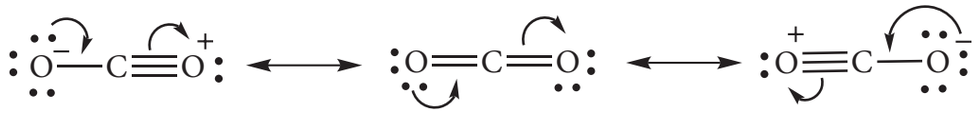


Figure 2.10 Structure of carbon dixide

Uses of carbon dioxide

1. Carbon dioxide is used to produce an inert atomosphere for chemical processing.
2. Biologically, it is important for photosynthesis.

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